

C Language Notes

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1. Introduction to C Language

C is a general-purpose programming language developed by Dennis Ritchie in 1972 at Bell Labs. It is fast, structured, and widely used for system software, operating systems, embedded systems, and application development.

2. Features of C

Main features of C are: simple syntax, fast execution, structured programming, portability, rich library support, pointers, memory management, and low-level access to hardware.

3. Basic Structure of a C Program

A C program mainly contains header files, main() function, variable declaration, statements, and return statement.

```
#include <stdio.h>
int main() {
    printf("Hello World");
    return 0;
}
```

4. Variables and Data Types

Variables are used to store data. Common data types are int, float, char, and double. Example: int age = 20; float marks = 85.5; char grade = 'A';

5. Operators in C

Operators are symbols used to perform operations. Types include arithmetic (+, -, *, /, %), relational (>, <, ==), logical (&&, ||, !), assignment (=), and increment/decrement (++/--).

6. Conditional Statements

Conditional statements are used for decision making. C supports if, if-else, else-if ladder, and switch statement.

7. Loops in C

Loops repeat a block of code. C supports for loop, while loop, and do-while loop. For loop is used when number of repetitions is known.

8. Arrays

Array is a collection of similar data types stored in continuous memory locations. Example: int marks[5] = {80, 70, 90, 60, 85};

9. Functions

Function is a block of code used to perform a specific task. It improves reusability and reduces code repetition. Example: `int add(int a, int b) { return a+b; }`

10. Pointers

Pointer is a variable that stores the address of another variable. Example: `int *p;` Pointers are useful in dynamic memory allocation, arrays, and functions.

11. Strings

String is a collection of characters ending with null character `'\0'`. Example: `char name[20] = "Sourabh";` Common string functions are `strlen()`, `strcpy()`, `strcat()`, `strcmp()`.

12. File Handling

File handling is used to create, read, write, and update files. Common functions are `fopen()`, `fprintf()`, `fscanf()`, and `fclose()`. Modes include `r`, `w`, `a`, `r+`, and `w+`.

13. Advantages of C

C is fast, powerful, portable, supports pointers, has rich libraries, and is useful for learning programming fundamentals.

14. Conclusion

C language is one of the most important programming languages. It helps students understand variables, loops, arrays, functions, pointers, and file handling.